Wetlands and prescribed chemistry in Peng et al. (2022)

Reference. Peng et al., *Nature* 612, 477–482, doi:10.1038/s41586-022-05447-w.

Method. The study combines updated bottom-up inventories, two wetland models, LMDZ–INCA chemistry, a halocarbon inversion, and a surface-methane inversion. The latter assimilates 103 stations with PYVAR–LMDZ–SACS at $1.9° \times 3.75°$, optimizing eight-day surface fluxes and initial concentrations. Gridded prior flux errors are 70%. OH is prescribed from LMDZ–INCA, rather than optimized jointly with methane emissions. The chemistry simulation uses ERA5 meteorology and CEDS emissions updated for 2020 using Carbon Monitor activity changes. The separate 12-box inversion of HFC-134a, HFC-32, and HCFC-141b provides an independent, prior-sensitive OH constraint. The wetland models ORCHIDEE–MICT and LPJ-wsl use four meteorological forcings; ORCHIDEE also varies its wetland-area calibration.

Findings. Relative to 2019, the published 2020 inversion emission anomaly is 6.9 ± 2.1 Tg CH_4 yr^{-1}, and the attributed OH sink reduction is 7.5 ± 0.8 Tg CH_4 yr^{-1}. These account for 47% and 53% of the reported growth-rate increase, respectively. Bottom-up anthropogenic emissions decrease by 1.2 ± 0.1 Tg yr^{-1} and fire emissions by 6.5 ± 0.1 Tg yr^{-1}, while modeled wetland emissions increase by 6.0 ± 2.3 Tg yr^{-1}. The inversion places a 14.3 Tg yr^{-1} increase in the Northern Hemisphere and a 7.4 Tg yr^{-1} decrease in the Southern Hemisphere. Chemistry gives a 1.6 ± 0.2% global OH decrease. The halocarbon estimate is 1.6–1.8% with 10% prior OH uncertainty, but only 0.6% with a 5% prior.

Supplement and interpretation. Supplementary Tables 1–2 document the meteorological differences and inversion priors. The soil sink is held at its 2017 climatology in 2019–2020, so a change in net surface flux can be compared with a gross-emission change if that prescribed uptake is retained. Supplementary Figs. 12–18 describe observational coverage and independent validation, including residual northern column and vertical-gradient biases. The lake-temperature calculation gives a small negative central anomaly but a broad range and excludes area and productivity changes; it does not constrain all aquatic emissions.

Use in this reconstruction. We acquired the archived monthly inversion netCDF and publisher source spreadsheets, and retain their numerical precision separately from rounded text values. Only 2019–2020 is available. The published 7.5 Tg sink contribution is retained as a reported attribution: a mean OH percentage alone does not establish the reaction-weighted loss frequency needed by He et al. We do not silently add a burden correction to this number or assign missing 2021–2022 values. A common-burden calculation is labeled as an additional assumption, rather than a reproduction of Peng's native budget.